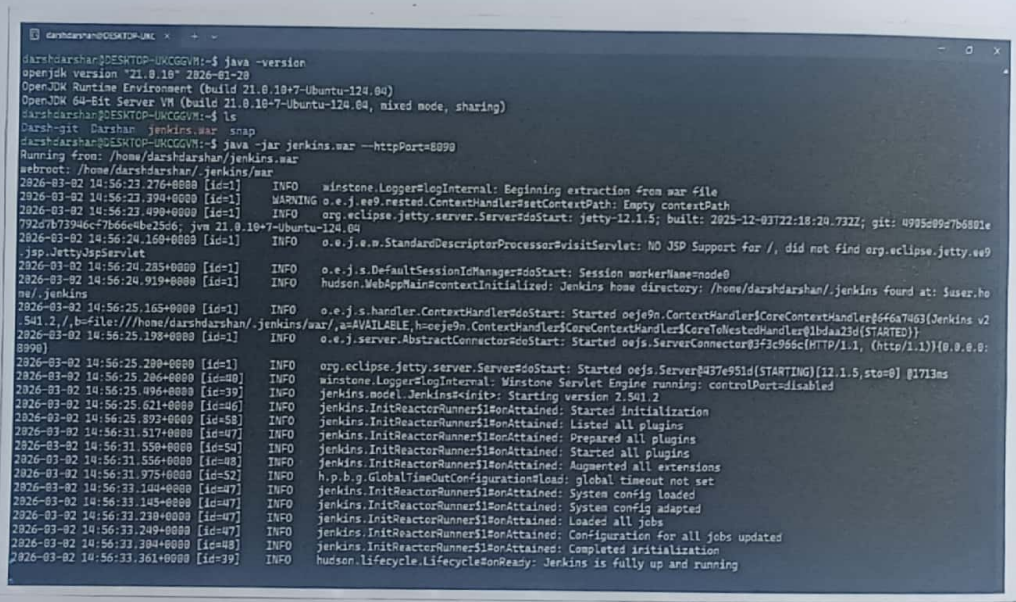

```
darshdarshar@DESKTOP-UMCGGVM:~$ java -version
openjdk version "21.0.10" 2026-01-20
OpenJDK Runtime Environment (build 21.0.10+7-Ubuntu-124.04)
OpenJDK 64-Bit Server VM (build 21.0.10+7-Ubuntu-124.04, mixed mode, sharing)
darshdarshar@DESKTOP-UMCGGVM:~$ ls
Darsh-git  Darshan  jenkins.war  snap
darshdarshar@DESKTOP-UMCGGVM:~$ java -jar jenkins.war --httpPort=8090
Running from: /home/darshdarshar/jenkins.war
webroot: /home/darshdarshar/.jenkins/war
2026-03-02 14:56:23.276+0000 [id=1]     INFO    winstone.Logger#logInternal: Beginning extraction from war file
2026-03-02 14:56:23.394+0000 [id=1]     WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-03-02 14:56:23.490+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: jetty-12.1.5; built: 2025-12-03T22:18:24.732Z; git: 4905c09d7b6801e
792d7b73946cf7b66e4be25d6; jvm 21.0.10+7-Ubuntu-124.04
2026-03-02 14:56:24.160+0000 [id=1]     INFO    o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9
.jsp.JettyJspServlet
2026-03-02 14:56:24.285+0000 [id=1]     INFO    o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2026-03-02 14:56:24.919+0000 [id=1]     INFO    hudson.WebAppMain#contextInitialized: Jenkins home directory: /home/darshdarshar/.jenkins found at: $user.ho
me/.jenkins
2026-03-02 14:56:25.165+0000 [id=1]     INFO    o.e.j.s.handler.ContextHandler#doStart: Started oej9n.ContextHandler$CoreContextHandler@646a7403{Jenkins v2
.541.2,/,b=file:///home/darshdarshar/.jenkins/war/,a=AVAILABLE,h=oej9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@1bdaa23d{STARTED}}
2026-03-02 14:56:25.198+0000 [id=1]     INFO    o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConnector@3f3c966c{HTTP/1.1, (http/1.1)}{0.0.0.0:
8090}
2026-03-02 14:56:25.200+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: Started oejs.Server@437e951d{STARTING}[12.1.5,sto=0] @1713ms
2026-03-02 14:56:25.206+0000 [id=40]    INFO    winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2026-03-02 14:56:25.496+0000 [id=39]    INFO    jenkins.model.Jenkins#<init>: Starting version 2.541.2
2026-03-02 14:56:25.621+0000 [id=46]    INFO    jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-03-02 14:56:25.893+0000 [id=58]    INFO    jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-03-02 14:56:31.517+0000 [id=47]    INFO    jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-03-02 14:56:31.550+0000 [id=54]    INFO    jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-03-02 14:56:31.556+0000 [id=48]    INFO    jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-03-02 14:56:31.975+0000 [id=52]    INFO    h.p.b.g.GlobalTimeOutConfiguration#load: global timeout not set
2026-03-02 14:56:33.144+0000 [id=47]    INFO    jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-03-02 14:56:33.145+0000 [id=47]    INFO    jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-03-02 14:56:33.238+0000 [id=47]    INFO    jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-03-02 14:56:33.249+0000 [id=47]    INFO    jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2026-03-02 14:56:33.304+0000 [id=48]    INFO    jenkins.InitReactorRunner$1#onAttained: Completed initialization
2026-03-02 14:56:33.361+0000 [id=39]    INFO    hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
```

Experiment No. 03 Date 03/03/2026 Page No 13

Name of Experiment Jenkins

Demonstrate the process of integration GitHub repository with Jenkins to automate the project execution in CI/CD pipeline.

sudo apt update

sudo apt install openjdk-21-jdk -y

ls

→ Check for jenkins.war file, if it exists, then

sudo rm jenkins.war
sudo rm -rf ~\.jenkins

→ If jenkins.war does not exists, then

wget https://get.jenkins.io/war-stable/latest/jenkins.war

→ Start Jenkins:

java -jar jenkins.war --httpPort=8097

→ Open the browser & access Jenkins using:

http://localhost:8097

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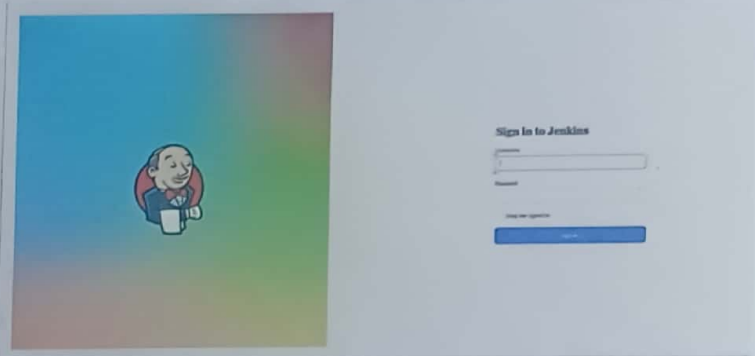

→ Copy the Administrator password shown in the terminal.

→ Paste it into the Jenkins unlock page

→ Click install Suggested Plugins.

Create First Admin User :

Enter the following details :-

Username : Akoshks05
Password : Akosh@2005
Fullname : Akosh K S
Email address : akoshksmysuru@gmail.com

Click Save & Continue → Save & Finish → Start Jenkins

GitHub Setup :

Create GitHub Repository :-

1. Open GitHub
2. Click New Repository
3. Enter repository name : simple-jenkins-demo
4. Click Create Repository.

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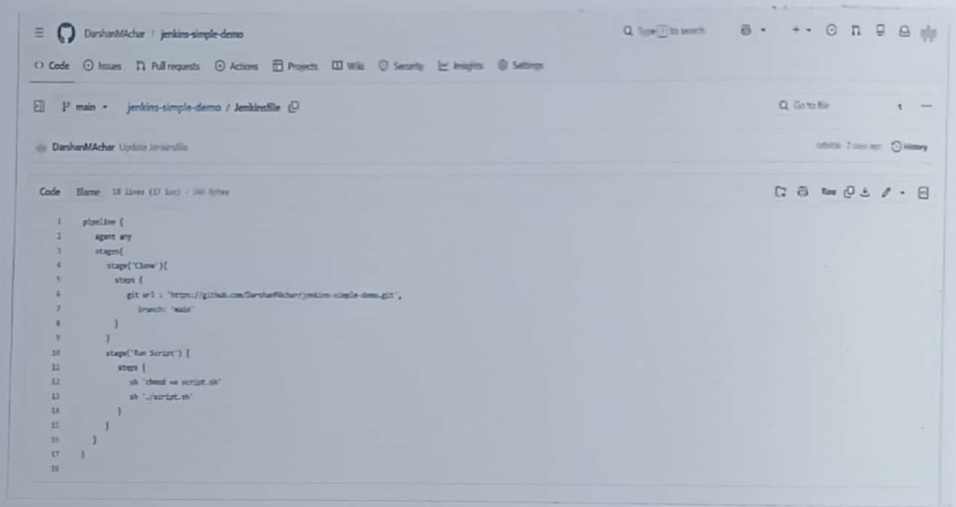

Experiment No. ..... Date ..... Page No 15

Name of Experiment .....

Add Files to Repository :

1. Click Add File → Create New file
2. Give file name Jenkinsfile
3. Add required script or code.
4. Click commit changes

Jenkinsfile :

```
pipeline {
    agent any
    stages {
        stage ('clone') {
            steps {
                git url : 'https://github.com/akashks05/
                            simple-jenkins-demo.git',
                    branch : 'main'
            }
        }

        stage ('Run Script') {
            steps {
                sh 'chmod +x script.sh'
                sh './script.sh'
            }
        }
    }
}
```

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Create another file:

1. File name : script.sh
2. Add shell script code.
3. Click Commit changes.

script.sh :

```
#!/bin/bash
echo "Hello from GitHub Project!"
echo "CI/CD Pipeline Executed Successfully!"
```

Jenkins Job Configuration:

Create New Jenkins Job:-

1. Open Jenkins Dashboard
2. Click New Item
3. Enter Job name : Akash
4. Select Pipeline
5. Click OK.

Configure Pipeline:-

1. Go to pipeline section → Triggers
2. Select Definition as Pipeline Script from SCM
3. Choose SCM → Git
4. Paste the GitHub repo URL

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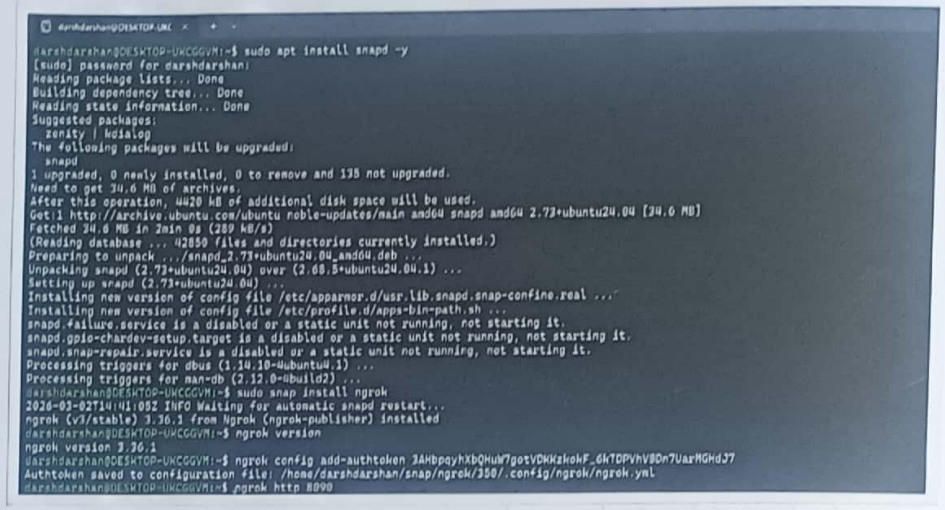
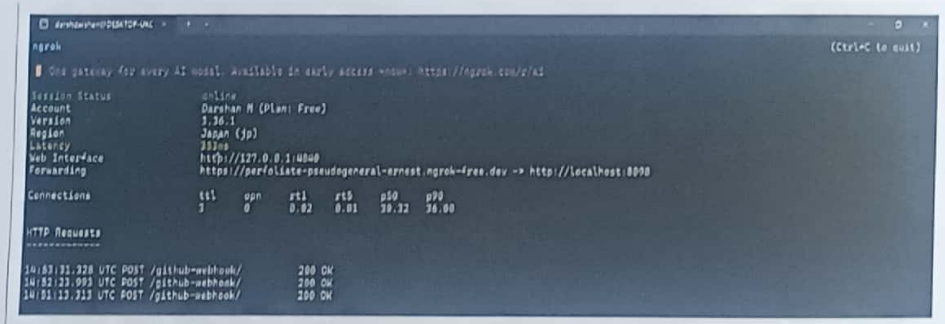


```
darshdarshan@DESKTOP-UKCGGVM: ~
darshdarshan@DESKTOP-UKCGGVM:~$ sudo apt install snapd -y
[sudo] password for darshdarshan:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  zenity | kdialog
The following packages will be upgraded:
  snapd
1 upgraded, 0 newly installed, 0 to remove and 135 not upgraded.
Need to get 34.6 MB of archives.
After this operation, 4420 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 snapd amd64 2.73+ubuntu24.04 [34.6 MB]
Fetched 34.6 MB in 2min 0s (289 kB/s)
(Reading database ... 42850 files and directories currently installed.)
Preparing to unpack .../snapd_2.73+ubuntu24.04_amd64.deb ...
Unpacking snapd (2.73+ubuntu24.04) over (2.68.5+ubuntu24.04.1) ...
Setting up snapd (2.73+ubuntu24.04) ...
Installing new version of config file /etc/apparmor.d/usr.lib.snapd.snap-confine.real ...
Installing new version of config file /etc/profile.d/apps-bin-path.sh ...
snapd.failure.service is a disabled or a static unit not running, not starting it.
snapd.gpio-chardev-setup.target is a disabled or a static unit not running, not starting it.
snapd.snap-repair.service is a disabled or a static unit not running, not starting it.
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
darshdarshan@DESKTOP-UKCGGVM:~$ sudo snap install ngrok
2026-03-02T14:41:05Z INFO Waiting for automatic snapd restart...
ngrok (v3/stable) 3.36.1 from Ngrok (ngrok-publisher) installed
darshdarshan@DESKTOP-UKCGGVM:~$ ngrok version
ngrok version 3.36.1
darshdarshan@DESKTOP-UKCGGVM:~$ ngrok config add-authtoken 3AHbpqyhXbQHuW7gotVDWKzkokF_6kTDPVhV8Dn7UarMGHdJ7
Authtoken saved to configuration file: /home/darshdarshan/snap/ngrok/350/.config/ngrok/ngrok.yml
darshdarshan@DESKTOP-UKCGGVM:~$ ngrok http 8090
```



```
darshdarshan@DESKTOP-UKCGGVM: ~
ngrok                                                    (Ctrl+C to quit)

 One gateway for every AI model. Available in early access mode: https://ngrok.com/r/ai

Session Status                online
Account                       Darshan M (Plan: Free)
Version                       3.36.1
Region                        Japan (jp)
Latency                       331ms
Web Interface                 http://127.0.0.1:4040
Forwarding                    https://perfoliate-pseudogeneral-ernest.ngrok-free.dev -> http://localhost:8090

Connections                   ttl     opn     rt1     rt5     p50     p90
                              3       0       0.02    0.01    30.32   36.00

HTTP Requests
-------------

14:32:31.328 UTC POST /github-webhook/         200 OK
14:32:23.993 UTC POST /github-webhook/         200 OK
14:31:13.313 UTC POST /github-webhook/         200 OK
```

Experiment No. ..... Date ..... Page No. 17

Name of Experiment .....

5. Change branch name to main
6. Click Save

Click on Build Now

Webhook Configuration (Automation):

Open another Ubuntu terminal,

ngrok --version

If ngrok is not installed, then
sudo apt update
sudo apt install snapd -y
sudo snap install ngrok

Login to ngrok account

1. Copy the auth token and config link from account & paste in new terminal.
2. ngrok http 8097

Copy the URL till .dev (ngrok-url)

Add Webhook in GitHub:

1. Go to GitHub Repository
2. Settings → Webhooks

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3. Click Add Webhook

Payload URL : ngrok-url/github-webhook/
Content type : application/json
Click Add Webhook.

Open Jenkins → Click Configure → Go to Triggers Section → Enable GitHub hook trigger for GitSCM polling

Click Save

Go to GitHub Repo
Add a file
Commit the changes

Jenkins will automatically trigger a build using the webhook.

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